5COSC020W Database Systems

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Part A

Questiom 1 : Critical Evalution

Strengths

1. The AI generated EERD indentifies store, product, member, and order as distinct entities.

- Every store sells a unique range of products and every member can place multiple orders, as in the business rules. Therefore this separation satisfies the principle of entity intergrity and normalisation.

2. The AI generated EERD effectively represents card payment and mobile payment as two specific types under the broader payment entity.

- while keeping common payment details like payment ID, amount and date in the main entity, it stores specific details such as card number or mobile number in the sub entities. This method helps to maintain data consistency and also it makes easier to add new paymen methods in the fututre.

3. The AI diagram properly shows a one – to – many relationship between member and order.

- Meaning, since each member can make multiple orders while each order belongs to one member. This matches the business rules of Foodz@Yourz and helps to maintain a clear and reliablr connection between customers and their orders.

Weaknesses

1. The AI EETD omits the shopping basket entity

- It is an important entity because members do their orders through baskets that can include products from multiple stores hence without this entity the model struggles to represent multi store selections. Therefore it reduces conceptual completeness.

2. Incomplete modelling of workforce and deliveries

- Even though Worker, Driver, or Collection are specifically mentioned in the project breif, the EERD fails to include them. These entities are crucial to record which employee handles or delivers an order, also how collections are organized. Their absence results to missing operational details and make it impossible to accurately track internal logistics.

3. Incomplete handling of product variations and perishability

- The AI generated model does not defferenciate products between perishable and non perishable, or even between soft drinks and alcoholic drinks. But according to the business rules, it requires the system to record validity dates for perishable items and store specific

Questiom :

2 Decision log

1. Created distinct entities for store, product, member, and order.

- The business rules explain that multi stores sell a wide range of products and each member can place several orders. Representing these entities as separate entities, which ensures entity intergrity and supports one – to – many relationships. This approach also fits with BQ1, which confirms that each store manages its own prices and stock independently.

2. Introduced a new entity called shopping basket.

- Members add and update items in their shopping baskets before placing an order. Introducing a Basket entity makes it possible to track selections from multiple stores and supports a one-tomany relationship between Member and Basket. This helps connect the process between product selection and order placement.

3. Established a many – to – many relationship between store and product through a stock detail associative entity.

- Each store can sell multiple products, and the same product can be appear in different stores with several prices and stock levels. Creating an associative entity allows attributes such as price and stock quantity to be stored per store product pair.

4. Represented products as a super entity with perishable and non perishable categories.

- The brief specifies that validity dates should be recorded only for perishable products. This justifies the use of specialisation to include attributes unique to perishable items, ensuring to maintain data accuracy and preventing null values in non-perishable records.

5. Created a sub-specialisation of Perishable Product into Soft Drink and Alcoholic Drink.

- The business differentiates between soft drinks and alcoholic beverages, each having unique attributes like alcohol percentage. Representing these as subtypes of Perishable Product follows the disjoint specialisation rule.

6. Defined Payment as a super-type with Card Payment and Mobile Payment as sub-types.

- Two payment methods are specifically mentioned in the brief. Using a supertype ensures common attributes are stored once, while subtypes hold unique details like card number or mobile number. This approach supports future flexibility.

7. Linked Order to Payment with a one-to-one relationship.

- Each order is linked to one payment transaction. This ensures referential integrity and simplifies financial reconciliation more straightforward.

8. Added entities for Employee Worker, Driver, and Collection.

- Operational staff and collection activities are key parts of Foodz@Yourz's logistics. The Collection entity groups several orders handled by one driver, enabling efficient order tracking and delivery management.

9. Introduced Vehicle as a sub-type entity (Two-Wheeler and Four-Wheeler).

- Drivers use two types of vehicles depending on the order volume. Representing Vehicle with subtypes helps represent different delivery capacities and ensures the system accurately reflects real-world logistics constraints.

10. Defined key relationships and multiplicities according to business logic.

- These relationships were identified from the business description, ensuring the EERD accurately represents real-world operations. So bacically each Member can have many Baskets, each Basket can include many Products, each Order is linked to one Payment, and each Collection can contain multiple Orders.

Questiom :

3 Entities

Entity Name	*Breif Explanation*
Store	This represents Foodz@Yourz each branch which is responsible for selling products and it's own stock levels and prices
Product	This shows general information about products offered by Foodz@Yourz, such as the name, description, etc.
Perishable	This is a subclass of product, so this represents items with limited shelf life that require expiry tracking.
NonPerishable	Even this is a subclass of product. In this it represents durable goods with no expiry constraints.
Food	This is a subclass of perishable. Which is for edible goods such as snacks, meals, etc.
Drink	This is a subclass of perishable. Which is for liquid consumables such as juice and alcohol.
SoftDrink	A subclass of drink, that represents non alcoholic beverages. For instance soda, juices and etc.
Alcoholic Beverages	A subclass of drink, that represents alcoholic beverages. For instance wine, beer and etc.
StockDetail	This links store and product to record stock quantity, unit pricr and also the availability in each store.
Basket	This is a cart created by the members temporarily before placing an order.
BasketItem	This connects Basket and Product entities to record quantity and date added for each selected product.
Order	Including the total amount, order date and order status this represents confirmed customer purchases.
OrderItem	Links Order and Product to record quantity and subtotal of each product in an order.
Payment	Stores payment transaction details linked to a specific oeder.
CardPayment	Stores details regarding card transactions, icluding card number and card type.

MobilePayment	Stores details regarding mobile or digital wallet transactions including mobile number and provider.
Employee	Represents all Foo@Yours staff members involved in order processing and delivery
Worker	Employees responsible for preparing and processing customer orders in the warehouse.
Driver	Employees responsible for delivering or collecting customer orders.
Vehicle	Represnts delivery vehicles used by drivers, it includes vehicle registration and capacity.
TwoWheeler	This includes motorbikes or scooters used for light deliveries.
FourWheeler	This includes vans or trucks used for heavy or bulk deliveries.
Collection	Represents the collection or delivery event grouping one or more customer orders.
Delivery	Connects Collection and Member to record delivery address, delivery status, and date.

Questiom :

4 Specialisations

General Entity	Specialised Entity	Breif Explanation
W2149487_Product	W2149487_Perishable W2149487_NonPerishable	Products are categorised as either non-perishable (longlasting) or perishable (having an expiration date). Since each product must be precisely one of these two categories, this specialisation is {disjoint, complete}.
W2149487_Perishable	W2149487_Food W2149487_Drink	Food and Drink are two more categories for perishable goods. Since all perishable goods are either edible or drinkable but not both, this is {disjoint, complete}.
W2149487_Drink	W2149487_SoftDrink W2149487_Alcoholic	Drinks are divided into two subcategories: alcoholic (Alcoholic) and non-alcoholic (SoftDrink). A drink cannot be both types, hence this is {disjoint, complete}.
W2149487_Payment	W2149487_MobilePayment W2149487_CardPayment	You can use a card or a mobile device to make payments. Since each payment utilises a single payment type, this is {disjoint, complete}.
W2149487_Employee	W2149487_Worker W2149487_Driver	Workers who prepare orders and drivers who deliver or pick them up are the two categories in which employees specialise. Since each employee fulfils a single, distinct duty, this is {disjoint, complete}.
	W2149487_TwoWheeler	Depending on their type, vehicles are classified as either four-wheelers for heavier loads or

W2149487_Vehicle	W2149487_FourWheeler	two-wheelers for lighter deliveries. Since each car only falls into one category, this is {disjoint, complete}.

5 Relationships & Multiplicities

Entitiy name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justification
W2149487_Store	1..*	Manages	1..*	W2149487_product	• Every store sells a variety of products. • Each product can be available in several stores. • A minimum of one product is required for stores to function. • Products may be present in the system but may not be available in every store.
W2149487_Member	1..1	Creates	0..*	W2149487_Basket	• Multiple baskets can be created by a member. • Each basket is owned exacly by one member. • Without a member, baskets cannot exist • Some members may have no baskets at a given time.
W2149487_Basket	1..1	Contains	0..*	W2149487_Product	• One or more products are included in each basket. • A product can be added to several baskets. • When a basket is checked out, it must contain at least one item. • A product may not currently be in any basket..

Questiom :

W2149487_Basket	0..*	Converts to	1..1	W2149487_Order	• When a basket is checked out, it becomes one order.

					• Every order comes from a single basket. • Some baskets might never be checked out. • Therefore, order side is optional, whereas basket side is mandatory.
W2149487_Order	1..1	Includes	1..1	W2149487_Product	• There are numerous items in each order. • Items may show up in more than one order. • At least one product must be included in each order. • Current orders might not include some products.
W2149487_Order	1..1	Paid by	0..*	W2149487_Payment	• One payment is required for each order. • Each payment refers to exactly one order. • A sale must be completed by both parties. • The 1-1 link prevents duplicate payments for each order.
W2149487_Worker	1..1	Works on	1..1	W2149487_Order	• A worker manages several orders. • One employee handles each order. • Orders may not always be processed by employees.

					• There is an optional order side and a one-to-many relationship.
W2149487_Driver	1..1	Assigned to	0..*	W2149487_Vehicle	• Each driver is assigned to one vehicle. • Each vehicle is operated by one driver.

					• Both sides are mandatory for accountability. • One-to-one ensures no overlap of driver-vehicle use.
W2149487_Driver	1..1	Delivers	0..*	W2149487_Collection	• A driver manages multiple collections. • Each collection is handled by one driver. • A driver must have at least one assigned collection. • Some collections may exist before driver assignment.
W2149487_Collection	1..1	Grouped	0..*	W2149487_Order	• Every collection consists of multiple orders. • Each order is part of a single collection. • At least one order must be included in a collection. • It's possible that certain orders aren't yet part of a collection.
W2149487_Collection	1..1	Delivered to	0..*	W2149487_Member	• Each collection/delivery is linked to a member. • A member can receive many deliveries. • Every delivery must have a recipient. • A member may have no current deliveries.
W2149487_Store	1..1	Employs	0..*	W2149487_Employee	• Each store employs multiple staff. • Each employee works at one store.

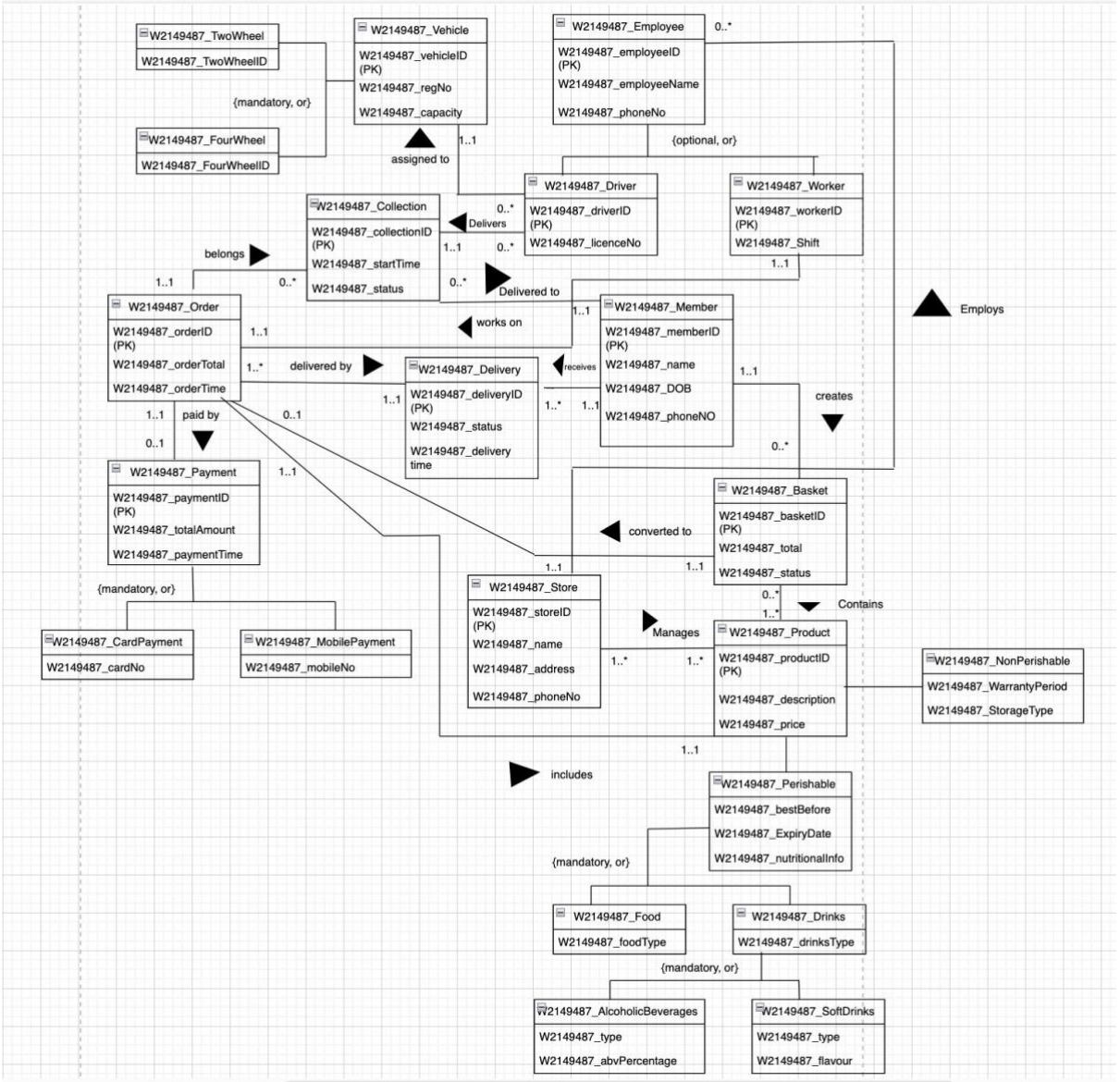
					• Stores must have staff to function. • Employees cannot exist without being assigned to a store.
W2149487_Member	1..*	Receives	1..1	W2149487_Delivery	• To be a member there should be minimum of one delivery. • Each delivery is linked to exactly one member. • This ensures that membership is meaningful and tied to actual deliveries. • A member can have multiple deliveries over time, reflecting repeat or ongoing service. •
W2149487_Order	1..1	Delivered by	1..*	W2149487_Delivery	• Each order must be delivered at least once. • A single order may involve multiple deliveries (e.g., split shipments). • Each delivery is associated with exactly one order. • This ensures that all orders are fulfilled and traceable through their deliveries.

Question 6: Attributes & Primary Key Data

Entitiy Name	Attributes for each entity	Brief explanation
Store	W2149487_storeID (PK) W2149487_name W2149487_address W2149487_phoneNo	Identifies each Foodz@Yourz branch and its address details.
Product	W2149487_productID (PK) W2149487_description W2149487_price	General details of each item sold by Foodz@Yourz.
Perishable	W2149487_productID (PK) W2149487_bestBefore W2149487_ExpiryDate W2149487_nutritionalInfo	Subclass of Product with expiry information.
NonPerishable	W2149487_ProductID (PK) W2149487_warrantyPeriod W2149487_StorageType	Subclass of Product for durable items with no expiry.
Food	W2149487_productID (PK, FK) W2149487_foodType	Subclass of Perishable for edible products.
Drink	W2149487_productID (PK, FK) W2149487_drinksType	Subclass of Perishable for liquid products.
SoftDrink	W2149487_productID (PK, FK) W2149487_type W2149487_flavour	Subclass of Drink representing nonalcoholic beverages.
Alcoholic	W2149487_ProductID (PK,FK) W2149487_abvPercentage W2149487_type	Subclass of Drink representing alcoholic beverages.
Basket	W2149487_basketID (PK) W2149487_total W2149487_status	Temporary shopping cart used by a member.
BasketItem	W2149487_basketItemID (PK) W2149487_ProductID W2149487_quantity	Connects Basket and Product; holds quantity added per basket.
Order	W2149487_orderID (PK) W2149487_orderTotal W2149487_orderTime	Records confirmed customer purchases.

Payment	W2149487_paymentID (PK) W2149487_totalAmount W2149487_paymentTime	Captures details of payments made for each order.
CardPayment	W2149487_cardNo	Subclass of Payment for card-based transactions.
MobilePayment	W2149487_mobileNo	Subclass of Payment for mobile wallet transactions.
Employee	W2149487_employeeID (PK) W2149487_employeeName W2149487_phoneNo	Represents staff working in Foodz@Yourz.
Worker	W2149487_workerID (PK) W2149487_Shift	Subclass of Employee responsible for preparing orders.
Driver	W2149487_driverID (PK) W2149487_licenceNo	Subclass of Employee responsible for deliveries.
Vehicle	W2149487_vehicleID (PK) W2149487_regNo W2149487_capacity	Represents vehicles used by drivers for deliveries.
TwoWheeler	W2149487_TwoWheelID	Subclass of Vehicle for motorbikes and scooters.
FourWheeler	W2149487_FourWheelID	Subclass of Vehicle for vans or trucks.
Collection	W2149487_collectionID (PK) W2149487_startTime W2149487_status	Represents grouped delivery or pickup events for orders.
Delivery	W2149487_status W2149487_delivery time W2149487_deliveryID (PK)	Records delivery details and status for each collection.

Question 7: Decision log


W2149487_TwoWheel
W2149487_TwoWheelID
W2149487_Vehicle
W2149487_vehicleID (PK)
W2149487_regNo
W2149487_capacity
W2149487_Employee
W2149487_employeeID (PK)
W2149487_employeeName
W2149487_phoneNo
0..*
{mandatory, or}
W2149487_FourWheel
W2149487_FourWheelID
assigned to
1..1
{optional, or}
W2149487_Driver
W2149487_driverID (PK)
W2149487_licenceNo
W2149487_Worker
W2149487_workerID (PK)
W2149487_Shift
W2149487_Collection
W2149487_collectionID (PK)
W2149487_startTime
W2149487_status
Delivers
0..*
1..1
0..*
belongs
1..1
0..*
0..*
Delivered to
W2149487_Order
W2149487_orderID (PK)
W2149487_orderTotal
W2149487_orderTime
1..1
1..1
works on
W2149487_Member
W2149487_memberID (PK)
W2149487_name
W2149487_DOB
W2149487_phoneNO
Employs
1..*
delivered by
W2149487_Delivery
W2149487_deliveryID (PK)
W2149487_status
W2149487_delivery time
receives
1..1
1..*
1..1
1..1
creates
1..1
paid by
0..1
0..1
0..*
W2149487_Payment
W2149487_paymentID (PK)
W2149487_totalAmount
W2149487_paymentTime
1..1
W2149487_Basket
W2149487_basketID (PK)
W2149487_total
W2149487_status
converted to
1..1
1..1
{mandatory, or}
W2149487_Store
W2149487_storeID (PK)
W2149487_name
W2149487_address
W2149487_phoneNo
0..*
1..*
Contains
Manages
W2149487_Product
W2149487_productID (PK)
W2149487_description
W2149487_price
W2149487_NonPerishable
W2149487_WarrantyPeriod
W2149487_StorageType
W2149487_CardPayment
W2149487_cardNo
W2149487_MobilePayment
W2149487_mobileNo
1..*
1..*
1..1
includes
W2149487_Perishable
W2149487_bestBefore
W2149487_ExpiryDate
W2149487_nutritionalInfo
{mandatory, or}
W2149487_Food
W2149487_foodType
W2149487_Drinks
W2149487_drinksType
{mandatory, or}
W2149487_AlcoholicBeverages
W2149487_type
W2149487_abvPercentage
W2149487_SoftDrinks
W2149487_type
W2149487_flavour

Business Questions

UOW ID :- W2149487

IIT ID :- 20241013

1. Can you confirm whether each store manages its own product prices and stock levels independently, like even for the same product?
2. When a member adds products from different stores into one basket, is this processed as a single combined order or as multiple store specific orders?
3. Would you like to identify differences between soft drinks and alcoholic beverages as separate categories, and if so, what additional details should be recorded for each?
4. Can one payment cover multiple orders, or is each order always linked to a single payment transaction?
5. Could you clarify whether warehouse workers handle one order at a time or several orders simultaneously?
6. Can a member create and manage multiple shopping baskets at the same time for different purposes like as an example for groceries and household items?

Assumptions

1. Member

- Each member has a unique MemberID.
- Members can place multiple baskets/orders, but each basket/order belongs to exactly one member.
- Contact details (phone, email) are optional.

2. Basket

- Each basket has a unique BasketID.
- A basket can contain multiple products via the BasketItem junction table.
- Each basket belongs to exactly one member.

3. Product

- Each product has a unique ProductID.
- A product can appear in multiple baskets via BasketItem.
- Product details like name, category, and price are stored.

4. BasketItem (Junction Table)

- Resolves the M:N relationship between baskets and products.
- Composite primary key: (BasketID, ProductID).
- Quantity is recorded for each product in the basket.

5. Order

- Each order has a unique OrderID.
- An order is linked to exactly one basket.
- Orders can have exactly one payment.

6. Payment

- Payment has a unique PaymentID.
- Each payment is linked to exactly one order.
- Payment details (amount, date, method) are stored.

7. Collection

- Each collection has a unique CollectionID.
- A collection can include multiple orders, reflecting orders collected together.

8. Vehicle & Driver (optional subtypes)

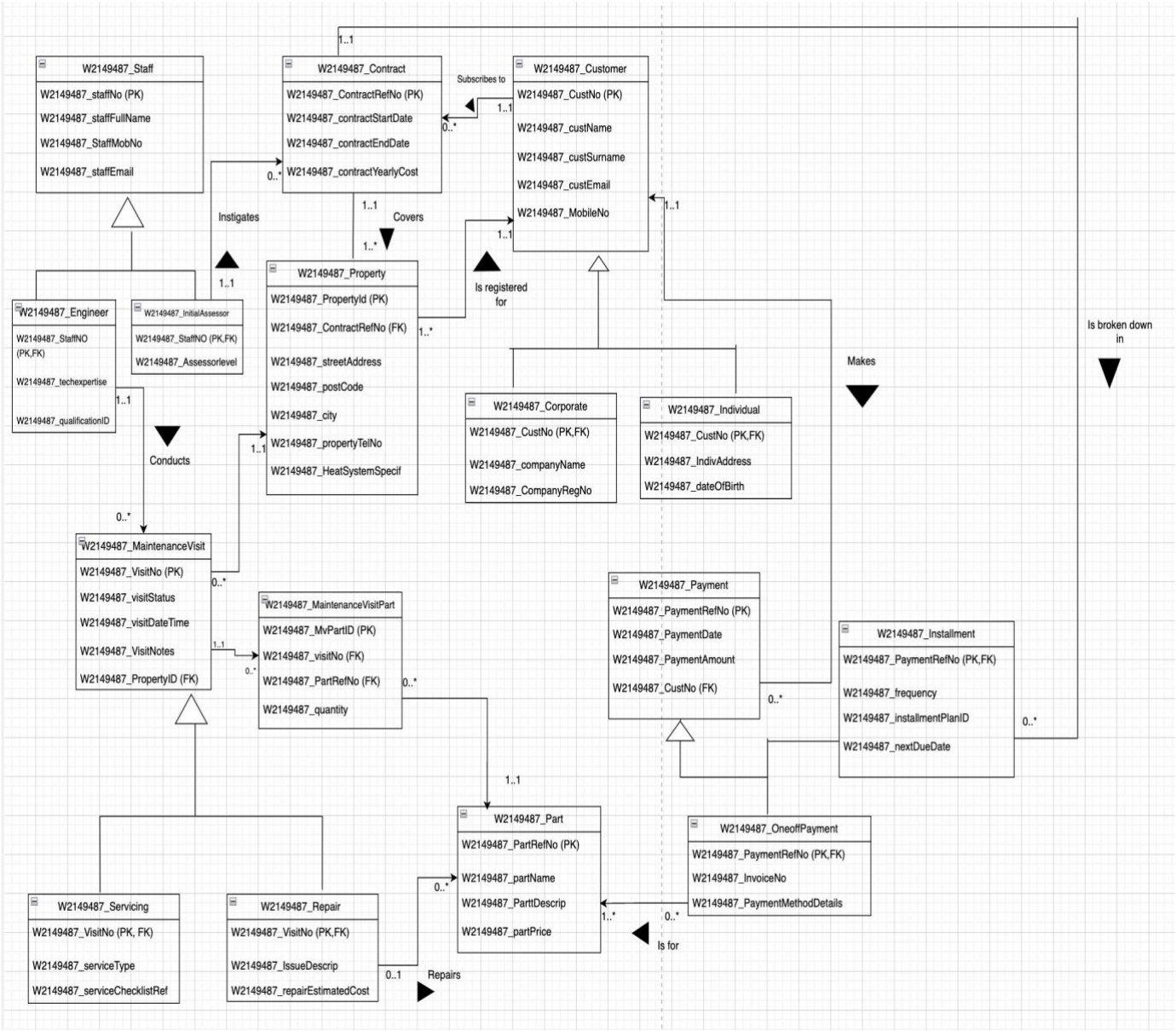
- Vehicles have subtypes depending on capacity.
- Drivers may operate different vehicles; each vehicle can be assigned to multiple collections.

9. General Assumptions

- Primary keys are unique and auto-generated.
- Foreign keys maintain referential integrity.

Part B

Questiom 8 : Mapping Conceptual EERD to Logical ERD



Questiom 9 : Mapping Log

I focused on making sure the conceptual model was correctly transformed into a useful relational structure when creating the logical ERD for BoilHeaterz. To prevent redundancy, I carefully assigned characteristics and defined primary keys while creating tables for each type of entity. In order to maintain inherited traits and specialization, I also kept subtypes like Corporate and Individual

Customers and Engineer and Initial Assessor personnel jobs, connecting them to their supertypes via primary–foreign key relationships.

I applied standard rules when mapping relationships: foreign keys in child tables were used to construct one-to-many links, such as Customers to Contracts and Contracts to Properties, reflecting the multiplicities in the business rules. The MaintenanceVisitPart associative table was used to resolve many-to-many relationships, especially between Maintenance Visits and Parts, capturing extra information like quantity. Similar to this, One – to – One PK/FK connections were used to map the MaintenanceVisit subclasses Servicing and Repair, enabling the system to record comprehensive visit information while preserving referential integrity.

In order to make sure the ERD reflects real-world procedures, such as properties having several maintenance visits or clients holding multiple contracts, I also paid attention to multiplicities and voluntary participation. In order to facilitate flexible connection to contracts or visits, payment management was organized using a supertype with subtypes for one-time and installment payments.

All things considered, developing this Logical ERD taught me how to strike a compromise between normalisation, referential integrity, and practical use, producing a design that is both technically sound and in line with BoilHeaterz's operational requirements.